Individual Pathways

Activity 4.2

Investigating chemical patterns

1. Insert a word to complete each sentence:

   (a) The total number of protons and neutrons is called the ..... number of an element.

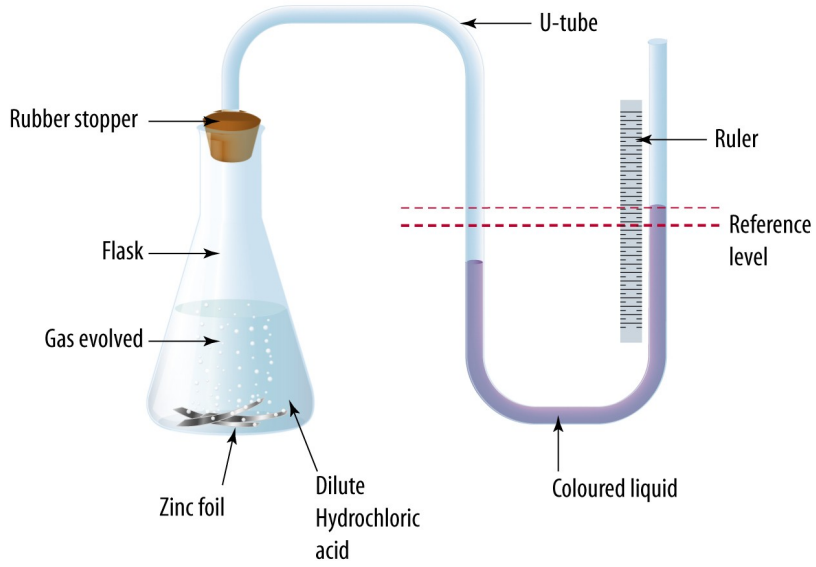
   (b) The attraction of a positive ion for a negative ions leads to the formation of an ..... bond.

   (c) Covalent bonds involve the sharing of ..... pairs.

   (d) The chemical formula of ..... is CH_4.

   (e) The chemical formula of copper (II) sulfate is.....

2. Examine the following apparatus.



   (a) Write a word equation for the chemical reaction occurring in the flask.

   .....
   .....

   (b) A coloured liquid is placed in the U-tube as shown. Initially the level of the coloured liquid in each arm of the U-tube is at the reference level line.

      (i) Why are the liquid levels initially equal?

      .....
      .....

      (ii) After some time, the level in the right arm has risen and the left arm has fallen as shown. Explain why this happens.

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.....
.....

(c) Explain how this apparatus can be used to follow the rate of the chemical reaction .....
.....
.....

(d) If magnesium was used instead of zinc, what differences would be observed?

.....
.....

3. Complete the following table. You may need to refer to the periodic table.

Element	Symbol	Electron configuration	Group	Period
Sodium	Na	2,8,1	1	3
Fluorine				
	P			
		2,2		
		2,8,7		
			18	4

4. The following diagrams show groups of four elements of the periodic table. For each grouping, state one property or characteristic that they all share.

(a)

9 Fluorine **F** **19.00**	10 Neon **Ne** **20.18**
17 Chlorine **Cl** **35.45**	18 Argon **Ar** **39.95**

.....

(b)

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97 Berkelium **Bk** **(247)**	98 Californium **Cf** **(251)**	99 Einsteinium **Es** **(254)**	100 Fermium **Fm** **(257)**

……..

(c)

73 Tantalum **Ta** **180.9**	74 Tungsten **W** **183.8**	75 Rhenium **Re** **186.2**	76 Osmium **Os** **190.2**

.....

5. Complete the following table by placing the **symbols** of the following elements in the correct columns.

Gold	Molybdenum	Tellurium	Potassium	Silicon
Germanium	Strontium	Lead	Iodine	Radon

METALS	METALLOIDS	NON-METALS

6. Draw the structural formulae for the following molecules.

(a) Methane (CH_4)

(b) Ammonia (NH_3)

7. Complete the table by writing the formulae of the ionic compounds that can form by combining ions of opposite charge:

	Ag^+	**Ba^{2+}**	**Fe^{3+}**
NO_3^-			
S^{2-}			
P^{3-}			

8. What is the valency of elements X, Y and Z in each of the compounds below?

   (a) H_3X (ionic) .....

   (b) YO_2 (covalent) .....

   (c) K_2Z (ionic).....

9. Using the provided information, determine the chemical formulae of each compound formed from each pair of elements.

   (a) R has an electrovalency of +3 and T has an electrovalency of –1.

   ……...

   (b) M has a covalency of 2 and Q has a covalency of 3.

   …….....

10. Using the provided information, determine the chemical formulae of each compound formed from each pair of elements.

    (a) D has an electron configuration of 2,8,18,8,2 and E has an electron configuration of 2,8,18,18,8,7.

    .....

    (b) G an electron configuration of 2,8,18,18,8,1 and H has an electron configuration of 2,8,18,6.

    .....

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